

1A

```
In [95]: import pandas as pd

data = pd.read_csv("Iris.csv")

data.head()

# df.describe()
```

```
Out[95]:
```

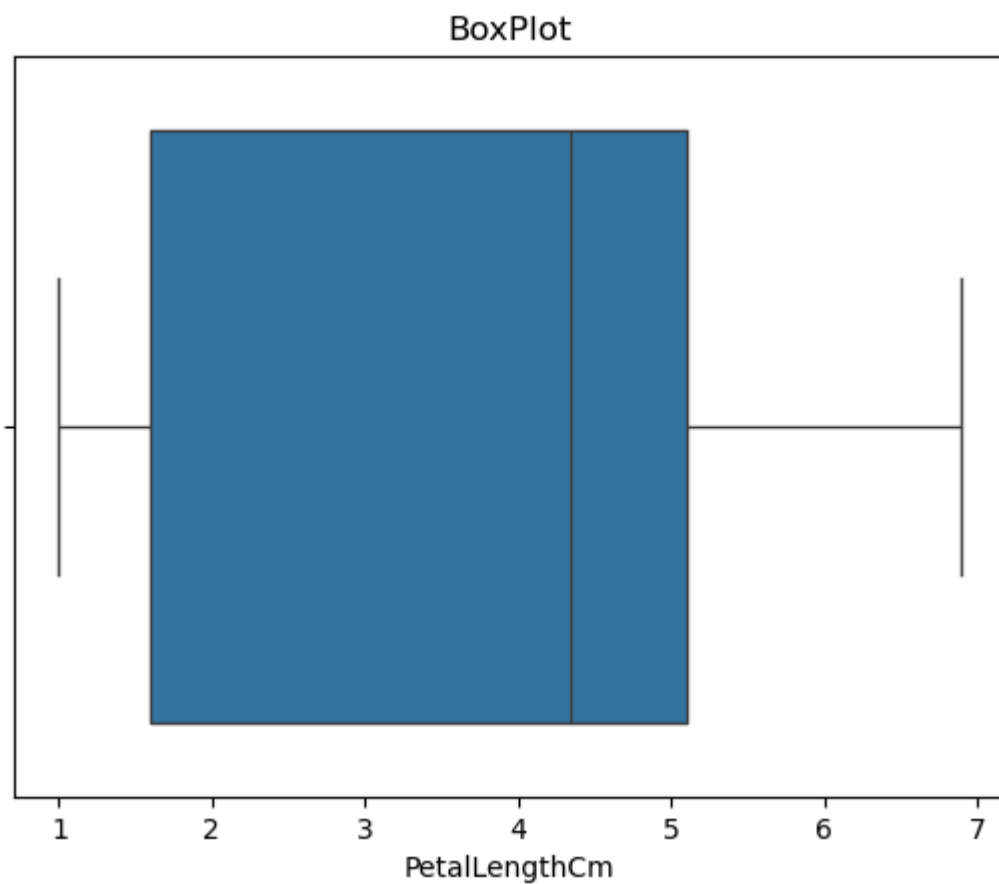
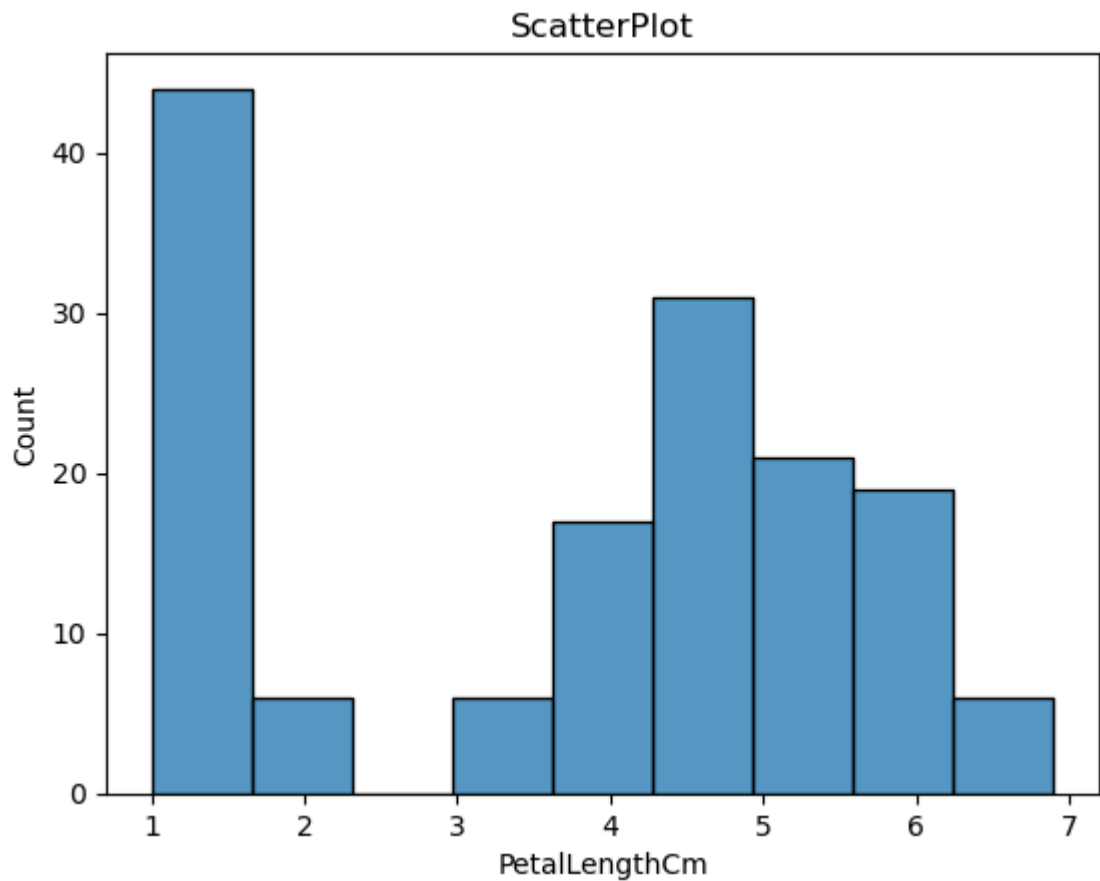
	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa

UNIVARIATE ANALYSIS

```
In [87]: import seaborn as sns
import matplotlib.pyplot as plt

sns.histplot(x='PetalLengthCm', data = data)
plt.title("ScatterPlot")
plt.show()

sns.boxplot(x='PetalLengthCm', data = data)
plt.title("BoxPlot")
plt.show()
```

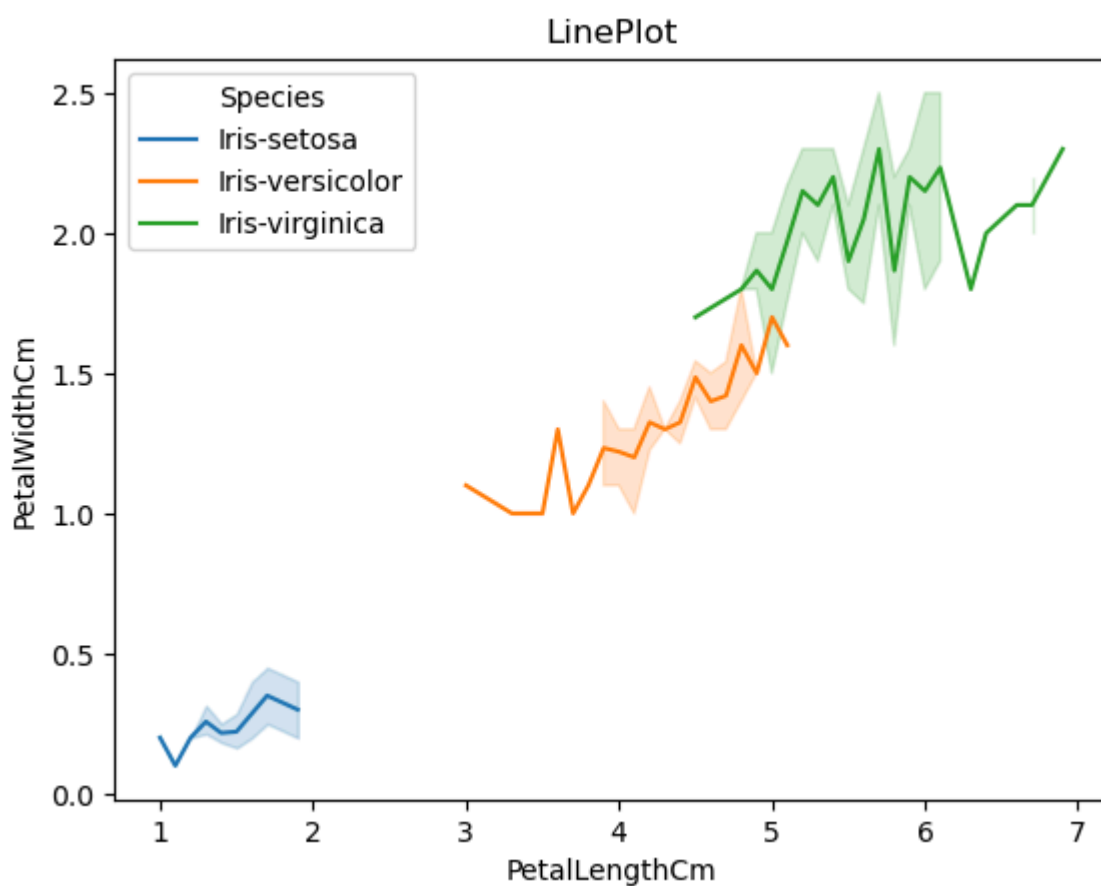
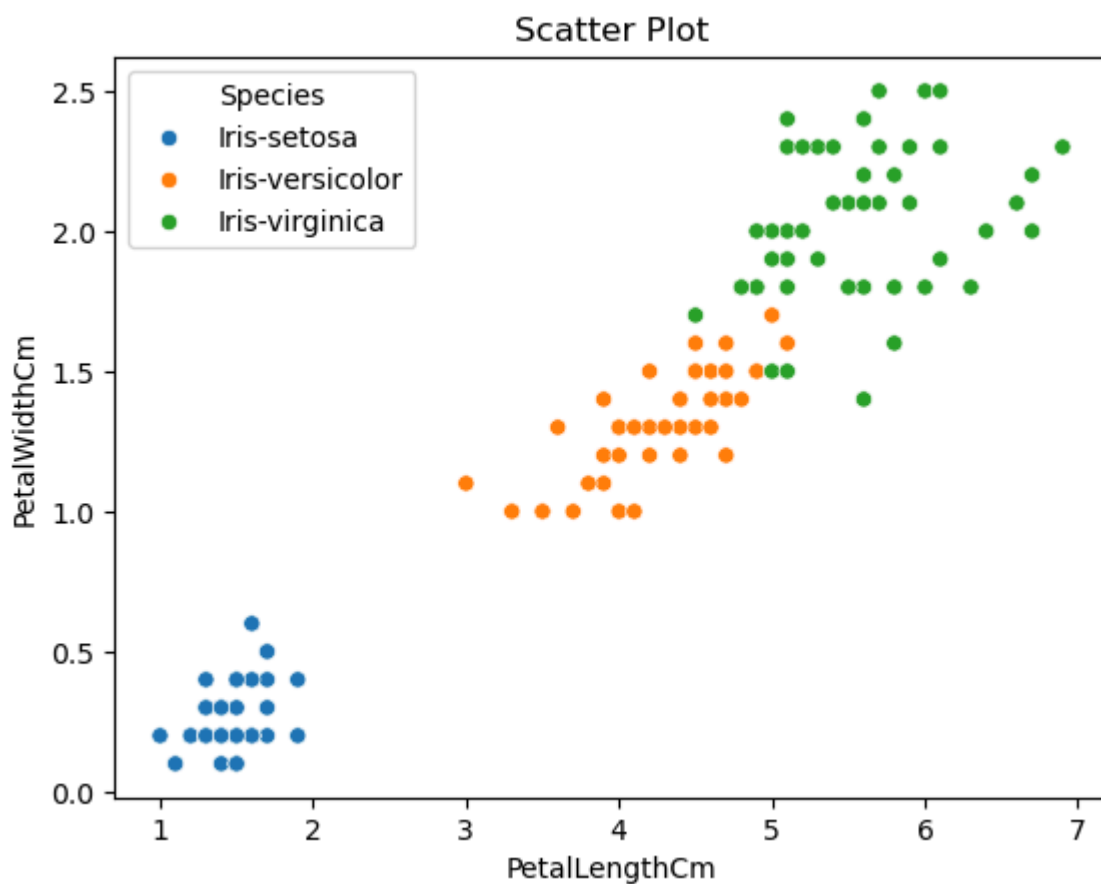


BIVARIATE ANALYSIS

```
In [105... sns.scatterplot(x='PetalLengthCm',y='PetalWidthCm',hue="Species",data=data)  
plt.title("Scatter Plot")
```

```
plt.show()

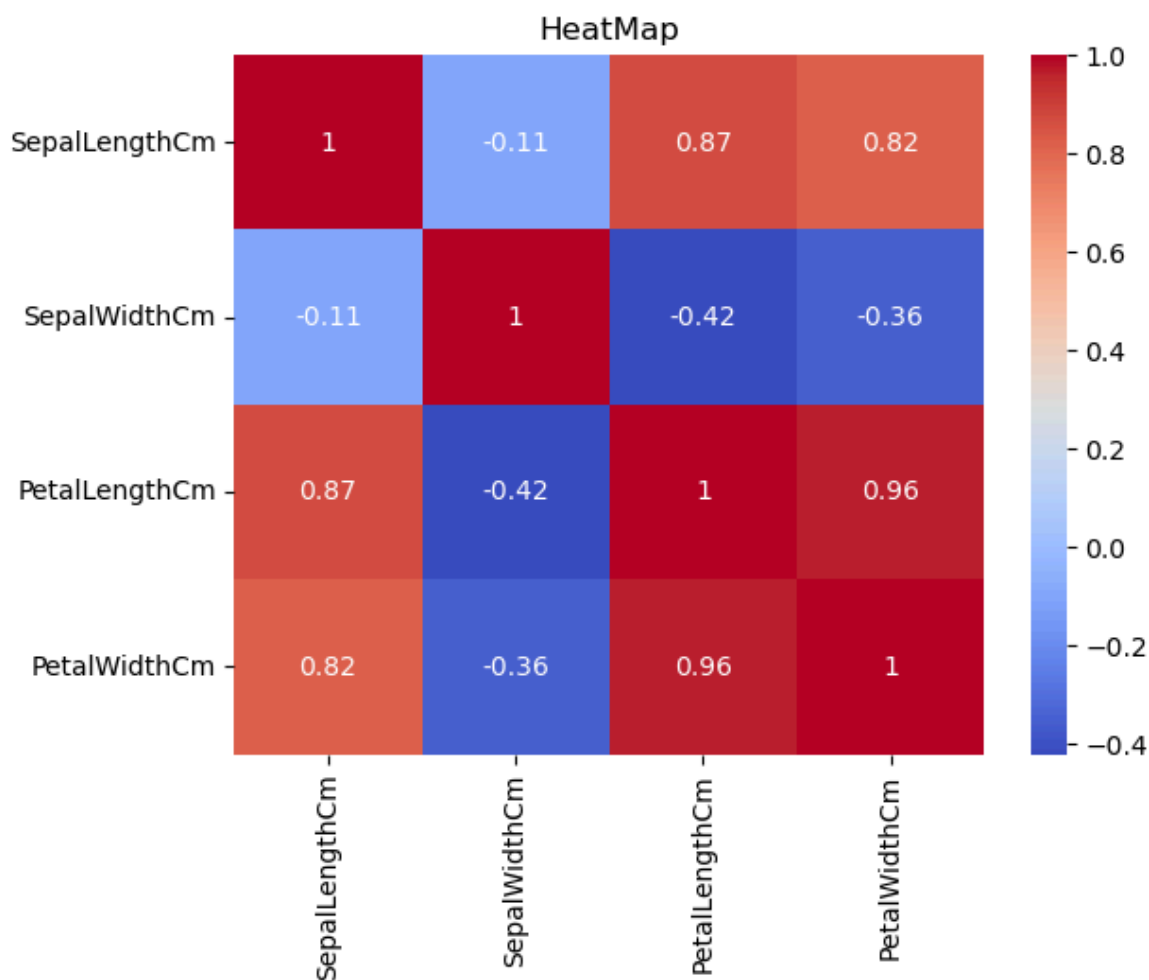
sns.lineplot(x='PetalLengthCm',y='PetalWidthCm',hue="Species",data=data)
plt.title("LinePlot")
plt.show()
```



MULTIVARIATE ANALYSIS

```
In [113... cols = ["SepalLengthCm", "SepalWidthCm", "PetalLengthCm", "PetalWidthCm"]
corr_matrix = data[cols].corr()

sns.heatmap(corr_matrix, annot=True, cmap='coolwarm')
plt.title("HeatMap")
plt.show()
```



1B

```
In [130... import pandas as pd

data = pd.read_csv("age.csv")

data.head()
```

Out[130...

	Age	Income	Years_Employed	Satisfaction
0	22	250000	1	6
1	25	320000	2	7
2	28	450000	5	8
3	30	500000	6	7
4	35	750000	10	9

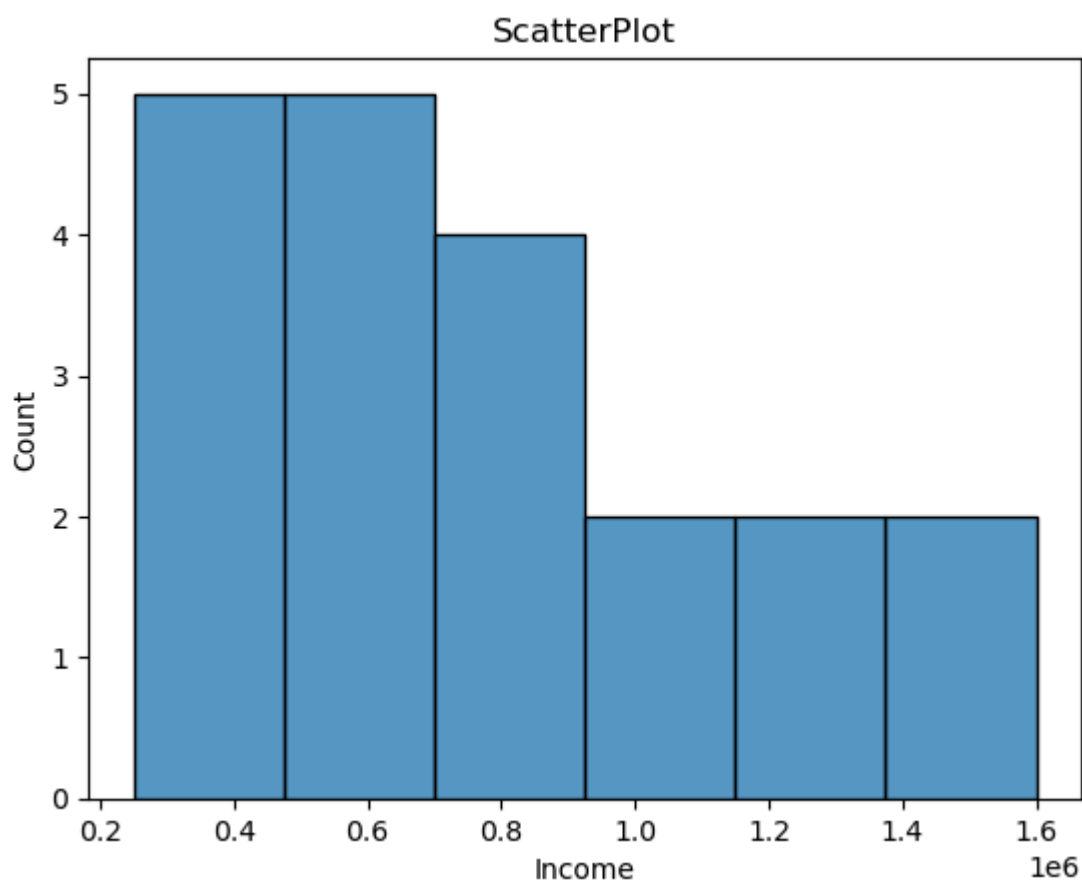
UNIVARIATE ANALYSIS

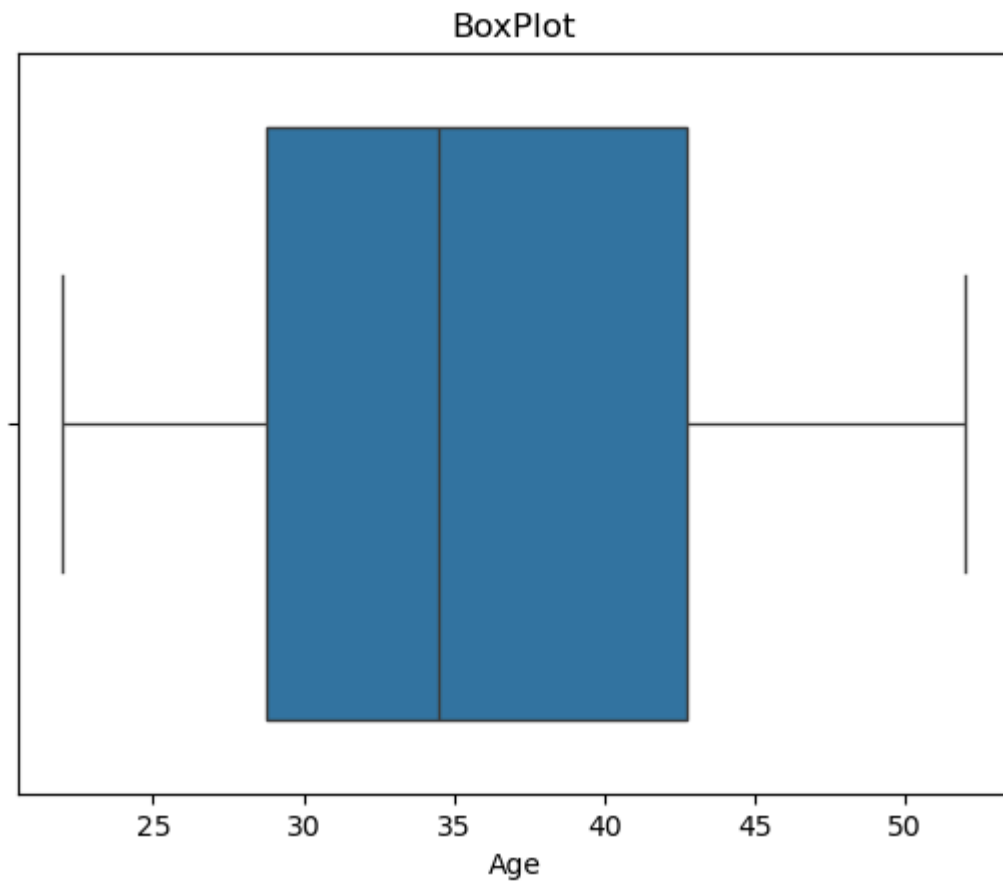
In [137...

```
import seaborn as sns
import matplotlib.pyplot as plt

sns.histplot(x='Income', data = data)
plt.title("ScatterPlot")
plt.show()

sns.boxplot(x='Age', data = data)
plt.title("BoxPlot")
plt.show()
```

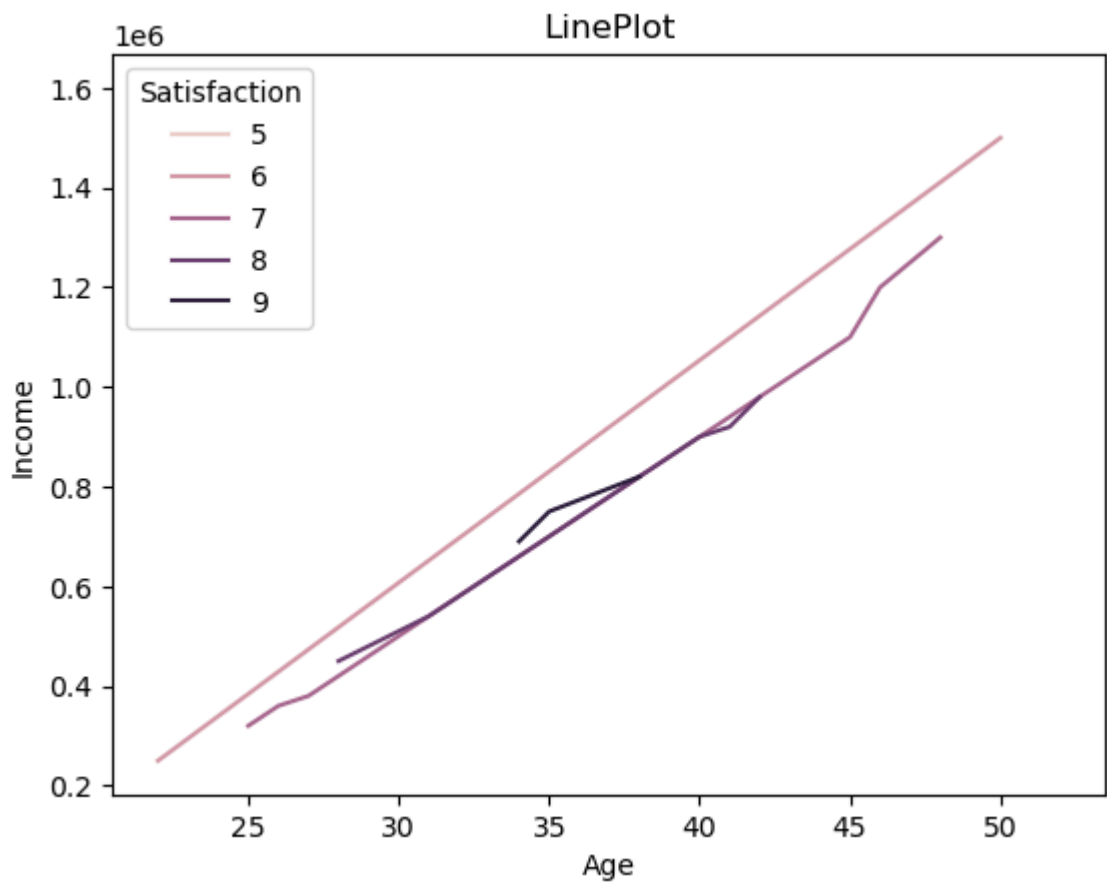
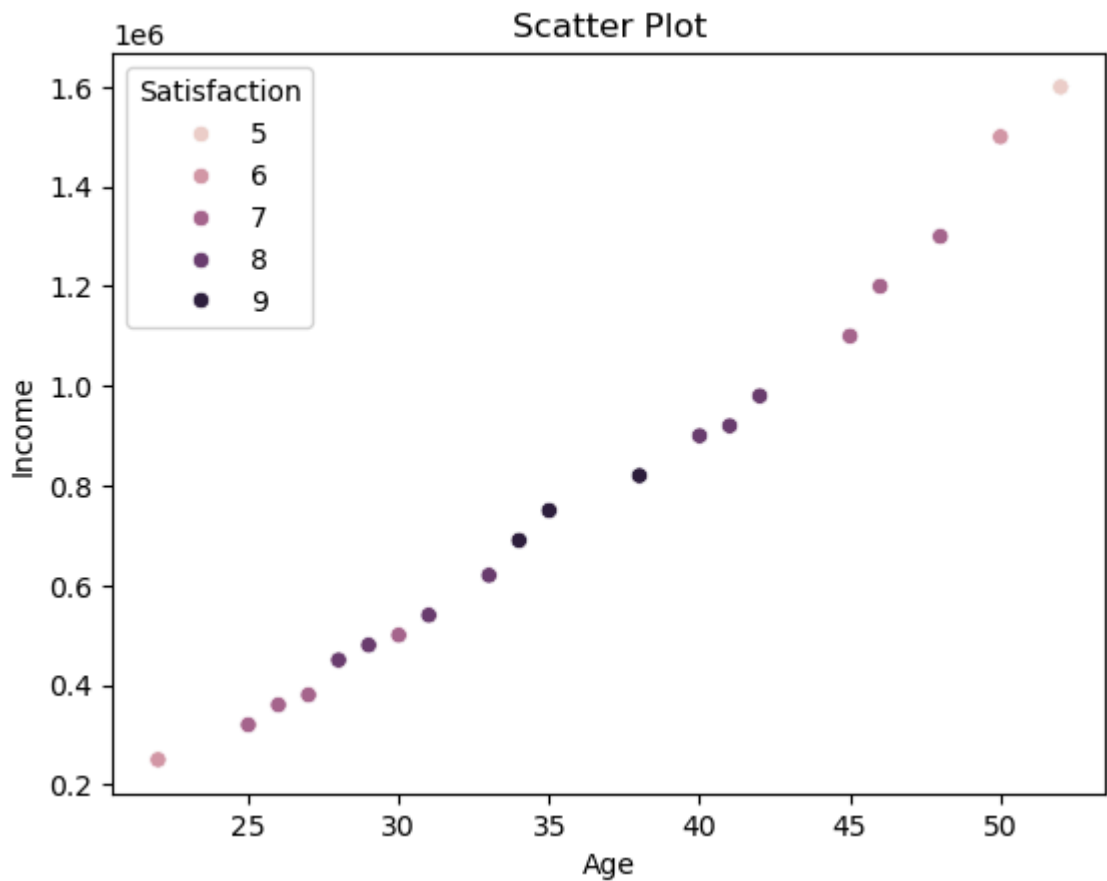




BIVARIATE ANALYSIS

```
In [143... sns.scatterplot(x='Age',y='Income', hue="Satisfaction", data=data)
plt.title("Scatter Plot")
plt.show()

sns.lineplot(x='Age',y='Income',hue="Satisfaction",data=data)
plt.title("LinePlot")
plt.show()
```



MULTIVARIATE ANALYSIS

```
In [145... cols = ["Age", "Income", "Years_Employed", "Satisfaction"]  
corr_matrix = data[cols].corr()
```

```
sns.heatmap(corr_matrix, annot=True, cmap='coolwarm')  
plt.title("HeatMap")  
plt.show()
```

